

Experiment 1B

Aim

File Sharing Within an Organization Using IP Address

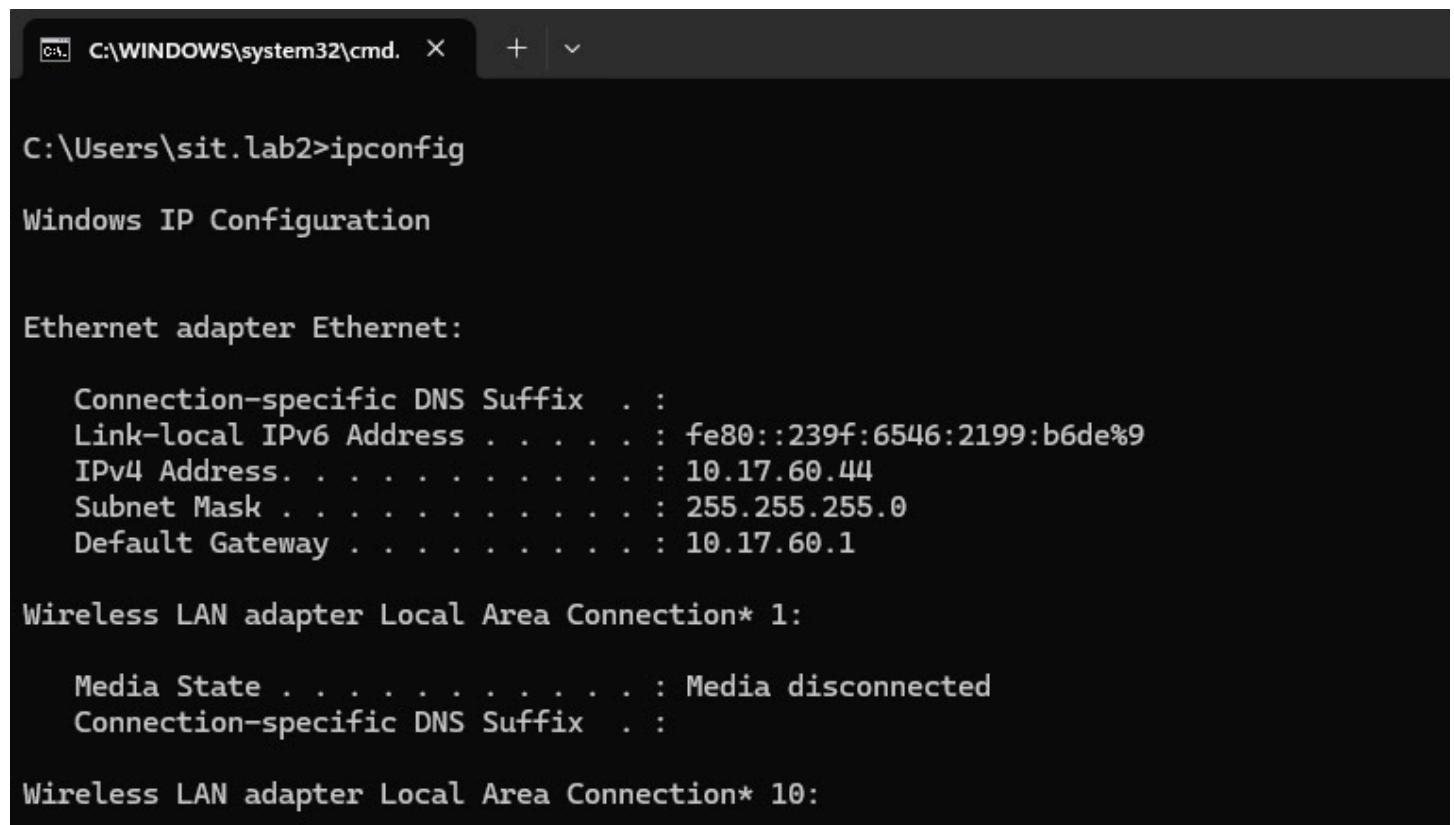
Theory

In a local area network (LAN), computers communicate with one another using unique numeric identifiers called **IP (Internet Protocol) addresses**. File sharing within an organization relies on the **Server Message Block (SMB)** protocol over TCP/IP, which allows client applications to read and write to files on a host server.

Steps Executed in Theory:

1. **Host Network Identification:** The host machine's IP address and network adapter details are retrieved using the network configuration command:

ipconfig



```
C:\WINDOWS\system32\cmd. X + v
C:\Users\sit.lab2>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::239f:6546:2199:b6de%9
    IPv4 Address. . . . . : 10.17.60.44
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.17.60.1

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:
```

2. **Network Connectivity Verification:** The connection stability and network latency between workstations are verified using the Internet Control Message Protocol (ICMP) echo request tool:

ping 10.17.60.44

[illegible]

3. **Directory Sharing Configuration:** A target directory is configured on the host machine (sit.lab2) with active read/write permissions enabled under **Properties -> Sharing -> Advanced Sharing**.
4. **Remote Access via Universal Naming Convention (UNC):** Client machines on the same subnet (255.255.255.0) connect directly to the shared resource over TCP/IP using the syntax:

\\10.17.60.44\Shared_Folder

Conclusion

The network interface configurations were successfully retrieved using the `ipconfig` command, and IP connectivity was verified using the `ping` utility with a 0% packet loss rate. File sharing within the organization was established and executed using the host system's IP address (\\10.17.60.44).